The maximum-entropy nature of black holes also has an important consequence for quantum gravity. In classical general relativity, there's nothing special about the interior region of a black hole, in between the event horizon and the singularity. There's a gravitational field there, but to an infalling observer it otherwise looks like empty space. According to the story we told in the last chapter, the quantum version of "empty space" is something like "a collection of spacetime degrees of freedom entangled together in such a way as to form an emergent three-dimensional geometry." Implicit in that description is that the degrees of freedom are scattered more or less uniformly throughout the volume of space we're looking at. And if that were true, the maximum-entropy state of that form would have all of those degrees of freedom entangled with the outside world. The entropy would thus be proportional to the volume of the region, not the area of its boundary. What's up?

There is a clue from the black hole information puzzle. The issue there was that there is no obvious way to transmit information from a book that has fallen into the black hole to the Hawking radiation emitted from the event horizon, at least not without signals moving faster than light. So what about this crazy idea: maybe all of the information about the state of the black hole—the "inside" as well as the horizon—can be thought of as living on the horizon itself, not buried in the interior. The black-hole state "lives," in some sense, on a two-dimensional surface, rather than being stretched across a three-dimensional volume.

First developed by Gerard 't Hooft and Leonard Susskind in the 1990s, based in part on a paper by Charles Thorn from 1978, this idea is known as the *holographic principle*. In an ordinary hologram, shining light on a two-dimensional surface reveals an apparently three-dimensional image. According to the holographic principle, the apparently three-dimensional interior of a black hole reflects information encoded on the two-dimensional surface of its event horizon. If this is true, maybe it's not so hard to get information from the black hole to its outgoing radiation, because the information was always on the horizon to start with.

Physicists still haven't settled on the precise meaning of holography for real-world black holes. Is it just a way of counting the number of degrees of freedom, or should we think that there is an actual two-dimensional theory living on the event horizon that describes the physics of the black hole? We don't know, but there is a different context in which holography is very